

Project Initialization and Planning Phase

Date	25 September 2026
Team ID	SWTID-2026-6721
Project Name	AI FAQ Assistant API
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

The AI FAQ Assistant project focuses on the difficulties faced by users and customer support teams when managing frequently asked questions. The system is planned to provide a centralized platform for creating, searching, managing, and generating FAQ content with the help of AI.

Identified Problem

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A customer support user managing frequently asked questions.	Find quick and accurate answers to common questions.	I have to search through many FAQ records manually.	FAQ information is difficult to organize and retrieve efficiently.	Frustrated and time-consuming.

PS-2	A content creator managing FAQ documentation.	Create and maintain useful FAQ content.	Writing answers and updating FAQs manually takes time.	There is no centralized AI-assisted FAQ management system.	Overwhelmed by repetitive work.
PS-3	An authenticated user.	Get an AI-generated answer to a question.	Preparing an appropriate answer manually takes additional effort.	The existing process does not provide automated AI assistance.	Delayed in getting a suitable answer.
PS-4	An administrator or FAQ manager.	Keep FAQ information secure and organized.	Unauthorized users may attempt to modify FAQ records.	FAQ management requires proper authentication and access control.	Concerned about data security and consistency.

From the above problems, our team identified the need for a **secure and AI-powered FAQ management system** that can:

- Manage frequently asked questions.
- Allow users to securely register and log in.
- Provide AI-generated answers.
- Generate FAQ question-answer pairs using Google Gemini AI.
- Search existing FAQ records.
- Allow authorized users to create, update, and delete FAQs.
- Store FAQ and user information securely using MongoDB.

Proposed Project

Our team proposed the **AI FAQ Assistant API**, a RESTful backend application developed using **Node.js**, **Express.js**, **MongoDB**, **Mongoose**, **JWT**, **bcrypt**, and **Google Gemini AI**.